

Multiple Choice (10 marks)

1. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
 - (a) ExoMars
 - (b) Mars Exploration Rovers
 - (c) Mars Science Laboratory
 - (d) Phoenix Mars Lander
2. The term Schwarzschild radius usually describes properties of ...
 - (a) red dwarfs.
 - (b) pulsars.
 - (c) black holes.
 - (d) galaxies.
3. How do astronomers think Jupiter generates its internal heat?
 - (a) nuclear fusion in the core
 - (b) by contracting changing gravitational potential energy into thermal energy
 - (c) internal friction due to its high rotation rate
 - (d) chemical processes
4. The nebular theory of the formation of the solar system successfully predicts all but one of the following. Which one does the theory not predict?
 - (a) Planets orbit around the Sun in nearly circular orbits in a flattened disk.
 - (b) the equal number of terrestrial and jovian planets
 - (c) the craters on the Moon
 - (d) asteroids Kuiper-belt comets and the Oort cloud
5. The famous Drake equation attempts to answer the following question:
 - (a) Will the Sun become a black hole?
 - (b) Is the universe infinitely large?
 - (c) How old is the visible universe?
 - (d) Are we alone in the universe?

True / False (10 marks)

Answer True or False

6. True or False: When a blackbody is cooled to half its original temperature, its power emitted decreases to 1/16 of its original value and its peak emission wavelength doubles.
7. True or False: Knowing both the actual brightness and apparent brightness of an object allows for the estimation of its distance from the observer.
8. True or False: Planet X is tidally locked, resulting in the absence of a solar day as it does not experience a full rotation relative to its sun.
9. True or False: Bolometric luminosity in astronomy is defined as the total luminosity integrated over all wavelengths of electromagnetic radiation.
10. True or False: Asteroids emit a considerable amount of their own radiation, which enhances their visibility in sky surveys.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the leading hypotheses regarding the origin of the Moon, and critically evaluate why the theory of the Moon splitting from the Earth due to tidal forces has not gained acceptance.
12. Discuss the significance of Betelgeuse within the Orion constellation, including its physical characteristics, cultural impact, and role in astronomical studies.

End of Paper